

Lab 3: Generative Discriminative Models

1 Generative Models

1.1 Load Unigram Model

- Load the unigram model from Lab 1 (Solutions are on Loop if you don't have your own implementation).
- Use the academy awards dataset from Lab 1 to train unigram model.

1.2 Generate Sample from Unigram Model

- Use the word probabilities created while training the unigram model to generate new samples.
- Hint: Use a random number generator

1.3 Plot different sample sizes

Plot the following word counts of samples with the following sizes: 100, 1000, 10000.

1.4 Plot the original word distribution

Plot the word distribution of the original academy awards dataset. What do you notice?

2 Discriminative Models

Use the scikit-learn implementation of Logistic Regression to train a model on the Nigerian English topic classification dataset that was used in Lab 2.